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Improving methane productivity of waste activated sludge by ultrasound and alkali pretreatment in microbial electrolysis cell and anaerobic digestion coupled system

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ABSTRACT

In order to enhance the productivity of methane from the waste activated sludge (WAS), a coupled system of microbial electrolysis cell (MEC) and anaerobic digestion (AD) was proposed. In this study, alkali, ultrasound-alkali, high-temperature coupled microaeration (TM) were applied as pretreatment methods to disintegrate the WAS flocs and break bacterial cell. After ultrasound-alkali pretreatment, the maximum accumulated concentration of VFAs and SCOD increased by 6.4 and 13.8 times compared with the initial concentration, which were 2.8 and 2.6 times of alkali pretreatment, and 2.1 and 2.1 times of TM pretreatment. Then, the pretreated sludge was transferred into MEC-AD coupled reactors and control group of AD reactors. The results showed that, methane production rate was enhanced to 0.15 m^3 CH_4/m^3 reactor/d in the coupled reactors, which was improved by 3 times compared with control AD (0.05 m^3 CH_4/m^3 reactor/d). The methane yield of MEC-AD coupled reactors achieved 808 ± 8 mL, which were increased by 97.0% ± 1.85% compared to control AD (410 mL). Using MEC can promote the rate of organics degradation and methane yield. The MEC-AD coupled system realized a good performance on the treatment of WAS and improved the efficiency of methane production.

1. Introduction

With the improvement of the living standard and the quality of human life, the amount of sewage and waste activated sludge (WAS) has been increased as well (Appels et al., 2008). As a result, the treatment and disposal of WAS have become a great challenge of environmental problems. The WAS is not a kind of waste (Jin et al., 2014). It has a relatively high biological value and economic value, because it contains large amounts of organic matters which has a high calorific value. Therefore, both developed and developing countries consider it as a significant renewable biomass energy (Wang et al., 2010). Many recent studies have involved the use of organic matters in sludge to produce new energy products. The studies of hydrogen and methane production are the most popular among them, and the two kinds of energy can meet the needs of industry (Lee et al., 2010). However, it still suffers from the low methanogenesis efficiency and poor stability, causing by the low hydrolysis rate of sludge and sensitivity of methanogens to the environment. Proper methods should be involved to improve the hydrolysis of sludge and enhance system stability. Studies have shown that some pretreatment methods can effectively promote sludge hydrolysis by releasing organic matters from the sludge flocs. In this way, the rate and yield of methane production were significantly increased. In addition, some coupling processes have also been coupled into the AD process to enhance strengths and avoid weaknesses of it. Among them, MEC-AD (anaerobic digestion) coupled reactor is mainly used to recycle WAS to produce methane.

Microbial electrolysis cell (MEC) is a kind of technology which can treat WAS while generating hydrogen from renewable biomass (Hu et al., 2009; Liu et al., 2012a; Logan, 2005; Rozendal et al., 2007; Schröder et al., 2015). MEC produces hydrogen faster, and hydrogen yield is also higher than other methods that utilize biomass, such as dark fermentation and photo-fermentation (Lu et al., 2012b). MEC can convert the complex organic substances in the sludge into hydrogen (Singh et al., 2013). The fermentation products of WAS, such as volatile fatty acids (VFAs), carbohydrates and proteins have been confirmed to be the substrate for hydrogen production (Wang et al., 2019). In MEC, the microbial community attached on the anode can use organic compounds, such as glucose and acetic acid (HAc) (Selembo et al., 2009,

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2010), which were oxidized to produce electrons and then the electrons will be delivered to the cathode. After the electrons reached the cathode, the electrons were combined with protons produced in anode to produce hydrogen under the anaerobic condition. Therefore, VFAs, carbohydrates and proteins can achieve cascade comprehensive utilization in MEC. MEC has become a versatile platform technology which provides a new solution for the environmental problems related to WAS disposal and energy resource recycle (Zhang and Angelidaki, 2014). Traditional AD methane production technology has been used to deal with WAS, which was an effective method for the conversion of organic matters to methane (Astals et al., 2013; Wahidunnabi and Eskicioglu, 2014). But AD has many shortcomings, such as low efficiency, low productivity, poor stability and so on (Jiang et al., 2010; Kargi et al., 2011; Xiao et al., 2011).

In this study, the potential technology improvement of AD was considered as the breakthrough point. It is attempted to introduce MEC into sludge treatment to realize sludge recycling by utilizing the advantages and characteristics of MEC (Pham et al., 2006; Villano et al., 2011, 2012). In the MEC-AD coupled reactor, a wide range of electrode microorganisms attached on anode can realize the substrate metabolisms in sludge to produce electrons for the aim of hydrogen production at cathode. This method can break through the limitation of slow rate of methanogenesis in anaerobic digestion process that was dominated by acetotrophic methanogens by promoting the hydrogenotrophic pathway. Hydrogen generated by MEC can stimulate hydrogenotrophic methanogens growth, and further promote the production of methane (Bo et al., 2014; Zhou et al., 2013). The addition of the MEC can improve conversion rate of the WAS and enhance the rate of methane production within a certain period of time. The combination of the MEC and AD can promote the production of methane, meanwhile, the degradation process of organic matters can be enhanced, and the production of the digested waste liquid can be further reduced (Guo et al., 2013). At the same time, the coupling of bioelectrochemistry process can quickly eliminate the inhibition factors and maintain the anaerobic environment of the system. The electrochemical effect can also eliminate the toxic effects of heavy metals on the system of various microorganisms (Luo et al., 2014; Nancharaiah et al., 2015).

The ways of WAS pretreatment technology have a significant influence on methane production (Ariunbaatar et al., 2014; Wang et al., 2017; Zhou et al., 2014a). As a general rule, WAS pretreatment is used to disintegrate the WAS flocs and break cell walls (Zhou et al., 2014b). In this study, three different pretreatment methods were used for pretreating the sludge, with high-temperature microaeration pretreatment, alkali pretreatment, and ultrasound-alkali pretreatment. The changes of soluble chemical oxygen demand (SCOD) and VFAs were usually used to measure the effect of the disintegration. Pretreatment method selection experiments were firstly conducted, and the most favorable pretreatment method was selected. The performance of the MEC-AD coupled system on improving methane productivity was evaluated, based on a series of batch experiments.

2. Materials and methods

2.1. Sludge properties

In this study, WAS collected from the secondary sedimentation tank of Taiping municipal wastewater treatment plant (Harbin City, Heilongjiang Province, China). The WAS firstly concentrated by gravity settling for 24 h, then the upper layer of the liquid was removed. Finally, the sludge was removed and filtered through 1 mm sieves to get rid of sand or other impurities, and its VSS was adjusted to 14000 mg L^{-1}. The main properties of sludge are listed in Table 1.

2.2. Reactor setup

MEC-AD coupled reactors were made of glass material, a cylindrical reactor with total volume of 690 mL (height: 18 cm; diameter: 7 cm), as shown in Fig. 1. According to the principle of AD and MEC, the reactor was divided into two reaction zones. The bottom was the AD reactor, and has a small rotor. The upper had an anode and cathode (Liang et al., 2011), anode was carbon brush (length: 0.8 cm; diameter: 0.4 cm; surface area: 1.01 m^2), and cathode was carbon cloth (3 cm in diameter) which was covered with a Pt catalyst layer on one side (0.5 mg Pt/cm^2). A power source (Switching Power Supply, FDPS-150, Fudantianxin Inc. China) was connected to the circuit of the reactor to add a fixed voltage of 0.8 V, and a data recorder was used to monitor the voltage across an external resistor (R = 10 Ω) for current calculating. The generated gas in the reactor was collected by the gas bag through a small opening at the top. The wall of the reactor on both sides had sampling port (diameter: 1 cm) for sampling. When the reactor was running, the openings were plugged with rubber plugs to form anaerobic environment. The size and material of the AD reactor were the same assembled as MEC-AD reactor, but with open circuit and no energy input.

Table 1
Main properties of raw sludge before adjustment.

Parameter	Average value[a]
pH	6.8 ± 0.1
TSS(mg/L)	24275 ± 256
VSS(mg/L)	15134 ± 124
SCOD (mg COD/L)	198 ± 10
TCOD (mg COD/L)	19252 ± 215
VFAs (mg COD/L)	21 ± 2
Solute carbohydrate (mg COD/L)	25 ± 3
Solute protein (mg COD/L)	156 ± 13

[a] All values are expressed in mg L^{-1} except pH.

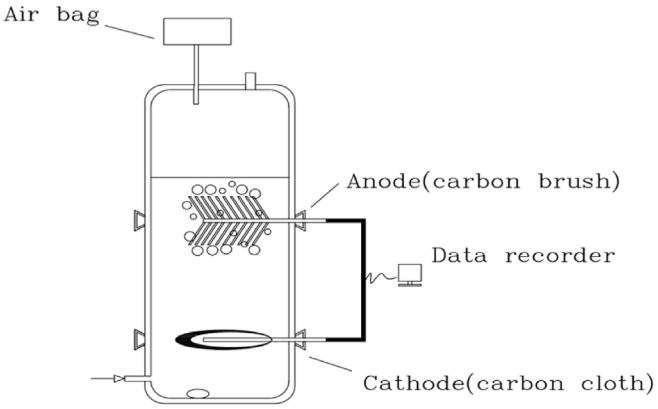


Fig. 1. MEC-AD coupled reactor.

2.3. Pretreatment methods

High-temperature microaeration pretreatment was conducted in water bath equipment (DZKW-S-4, Yongguangming Inc., Beijing). The temperature inside WAS kept at 55 °C by water-bath heating. The dissolved oxygen concentration was controlled by the air flow meter to be less than 0.5 mg L^{-1}. For alkali pretreatment, NaOH solution (6 mol L^{-1}) was added into the sludge for adjusting pH value. The pH value of the sludge was adjusted to pH 10. After 5 min of settling, the pH was measured and adjusted to pH 10. Settled for 5 min and repeated again.

For ultrasound-alkali pretreatment, sludge was firstly pretreated by ultrasound and then pretreated with alkali (Jin et al., 2009). Ultrasound pretreatment was conducted with a double-frequency ultrasonic instrument (KQ-600VDV, Kunshan Shumei Inc., China) at a frequency of 24 + 48 KH_Z. The ultrasound power density was 0.5 kW L^{-1}, and the pretreatment time was 10 min. After the ultrasound pretreatment, the

pH was adjusted to pH 10 by NaOH solution. After 5 min of settling, the pH was measured and adjusted to pH 10. Settled for 5 min and repeated again.

2.4. Batch experiments

WAS pretreatment experiments were conducted in serum bottles. Three copies of 500 mL WAS were pretreated by high-temperature microaeration, ultrasound-alkali, and alkali, respectively, and fermented in a shaking bath for 120 h at 35 °C before entering into MEC-AD reactor. The variations in VFAs, carbohydrates, proteins and SCOD contents were measured in a fixed time. Analyzing and comparing the effects of the sludge disintegration in three pretreatment methods, and selecting the most favorable pretreatment method to conduct the follow-up experiment.

The experiments of methane production were conducted in three reactors (M1, M2 MEC-AD coupled reactor and AD reactor). Reactor M1 and M2 have the same configuration with built-in electrodes discussed before. For the previous 16 days, the operation mode of M1 and M2 was the same. On the 16th day, the M2 reactor was replaced with a new electrode. AD reactor was the control group, and no electrochemical device was added. The pretreated WAS put in the all three reactors for anaerobic fermentation at room temperature (20–25 °C) for 21 days. Reactors M1 and M2 were operated at the added voltages of 0.8 V (Liu et al., 2012b; Xu et al., 2013), and carbon brush anode was put into the reactors on fifth day. Samples were taken every six days. The variations in methane yield, VFAs, proteins, carbohydrates, ammonia nitrogen, phosphorus, SCOD, VSS, and TCOD were measured and recorded.

2.5. Analysis and calculation

Gas chromatograph (9790II, Shanghai Analytical Apparatus, China) was used to analyze the gas composition (H_2, CH_4 and CO_2), with a packed column (TDX-01; 2 m length) and a TCD detector. VFAs mainly include six species, namely acetic (HAc), propionic (HPr), *n*-butyric (*n*-HBu), iso-butyric (iso-HBu), *n*-valeric (*n*-HAa) and iso-valeric (iso-HAa) acids. VFAs were measured in another gas chromatograph (Agilent, Technologies, Inc., USA), equipped with an Agilent Hi-Plex H column (300.0 × 6.5 mm) and a refractive index detector. Total suspended solids (TSS), volatile suspended solids (VSS), total chemical oxygen demand (TCOD) and SCOD were measured according to the standard methods. Proteins were measured with Lowry' method using bovine serum albumin as a standard solution. Carbohydrates were measured with phenol-sulfuric acid method using glucose as a standard solution. The concentration of ammonia nitrogen and phosphorus were measured by Nessler's reagent colorimetry and aluminum potassium sulfate degradation method, respectively. The current and voltage during fermentation was recorded by a data recorder.

SCOD removal efficiency is an essential performance index of sludge treatment, and it is also the most intuitive indicator. In MEC, SCOD removal efficiency was calculated according to the difference of SCOD value between the influent and effluent. SCOD removal efficiency was calculated according to equation (1):

$$SCODR = \frac{CODin - CODout}{CODin} \times 100\% \quad (1)$$

where COD_{in} (mg L^{-1}) is influent concentration of COD, COD_{out} (mg L^{-1}) is effluent concentration of COD, $SCOD_R$ (%) is SCOD removal efficiency.

3. Results and discussion

3.1. Effect of pretreatment on WAS disintegration

3.1.1. High-temperature microaeration pretreatment

As shown in Fig. 2(a), VFAs of the sludge got cumulated after high-temperature microaeration pretreatment. Within the first 24 h, the accumulation of VFAs was rapidly increased from less than 400 mg L^{-1} to 1190 mg L^{-1}, increased by about 3 times. At the moment, the concentration of HAc was 725 mg L^{-1} which accounted for 60.9% of the VFAs and the concentration of HPr was 110 mg L^{-1} which accounted for 9.2%. Besides, the concentration of *n*-HBu, iso-HBu, *n*-HAa and iso-HAa were 58, 95, 18 and 184 mg L^{-1}, respectively. HAc was the main production in the process of acidification, and it also was the main substrate that MEC utilized (Wang et al., 2014). The maximum concentration of VFAs reached 1915 mg L^{-1}. At the same time, ultimate carbohydrate concentration increased from the initial 59–298 mg L^{-1}. The concentration of protein reached 949 mg L^{-1}. In addition, the concentration of SCOD increased from 546 to 3333 mg L^{-1}, increased by 6.1 times. The increase of SCOD was mainly due to the disintegration of WAS and the release of organic compounds. Therefore, the changes of SCOD can be used to characterize the effects of pretreatment on WAS broken.

3.1.2. Ultrasound-alkali pretreatment

The VFAs accumulation after ultrasound-alkali pretreatment was shown in Fig. 2(b). VFAs were also rapidly increased from 480 to 3049 mg L^{-1} in 0–24 h. And compared to the initial values, it was increased by 6.4 times. After 24 h, VFAs concentration was continued to increase, but the rate of increase became slower gradually. After 100 h, VFAs concentration became stable, its maximum concentration was 4052 mg L^{-1}. HAc was the major accumulation product, which was accounted for 51.8% of total VFAs. The second was HPr, accounted for 23.4%, and others had no obvious changes. HAc increased fastest in 24 h from 235 to 1580 mg L^{-1} in the first 24 h, and then it gradually increased to the maximum concentration of 2054 mg L^{-1} at last. The trend of other acids was similar to HAc, the maximum concentrations of HPr and other acids were 919 mg L^{-1} and 300 ± 50 mg L^{-1}, respectively. Total carbohydrates could reach 320 mg L^{-1} within 24 h, and it steadily decreased to the final concentration of 134 mg L^{-1}. In addition, the protein concentration decreased from 1522 to 274 mg L^{-1}. The hydrolysis of soluble protein and carbohydrate was the main reason for the VFAs and SCOD accumulation. The concentration of SCOD increased from 498 to 6872 mg L^{-1}, increased by 13.8 times.

3.1.3. Alkali pretreatment

As shown in Fig. 2(c), VFAs concentration increased from 395 to 1428 mg L^{-1} after alkali pretreatment. The concentration of HAc increased from 201 to 676 mg L^{-1}, which was accounted for 47.3% of VFAs. HPr increased from 105 to 337 mg L^{-1}, and it was accounted for 23.6% of VFAs. After 24 h of fermentation, VFAs showed a slight decline. The overall concentration of VFAs was maintained within the range of 1400 ± 32 mg L^{-1}. HAc also decreased, the reason was that a small amount of HAc was consumed as the substrate by methanogens to produce methane. Whereas, the concentration of HPr increased from 337 to 424 mg L^{-1}, and that of other acids were not changed significantly. After 120 h of acidogenic fermentation, the carbohydrate concentration increased from 97 to 125 mg L^{-1}. Protein concentration increased from 133 to 380 mg L^{-1} after 24 h, but it remained at about 250 mg L^{-1} after 96 h. Further analysis showed that the hydrolysis of a large amount of proteins was the main reason for the accumulation of HAc after 24 h. SCOD concentration increased from 240 to 2667 mg L^{-1}, increased by 11.1 times.

3.1.4. Select the favorable pretreatment method

According to the experimental results (Table 2), it can be concluded that the pretreatment method of ultrasound-alkali has a better effect on disintegrating WAS. The quantity of dissolved matters of sludge pretreated with ultrasound-alkali was much higher than that of other pretreatment methods. Compared to the initial concentration, SCOD concentration in the WAS that was pretreated by ultrasound-alkali can be increased by more than 13 times. By contrast, that of alkali and high-

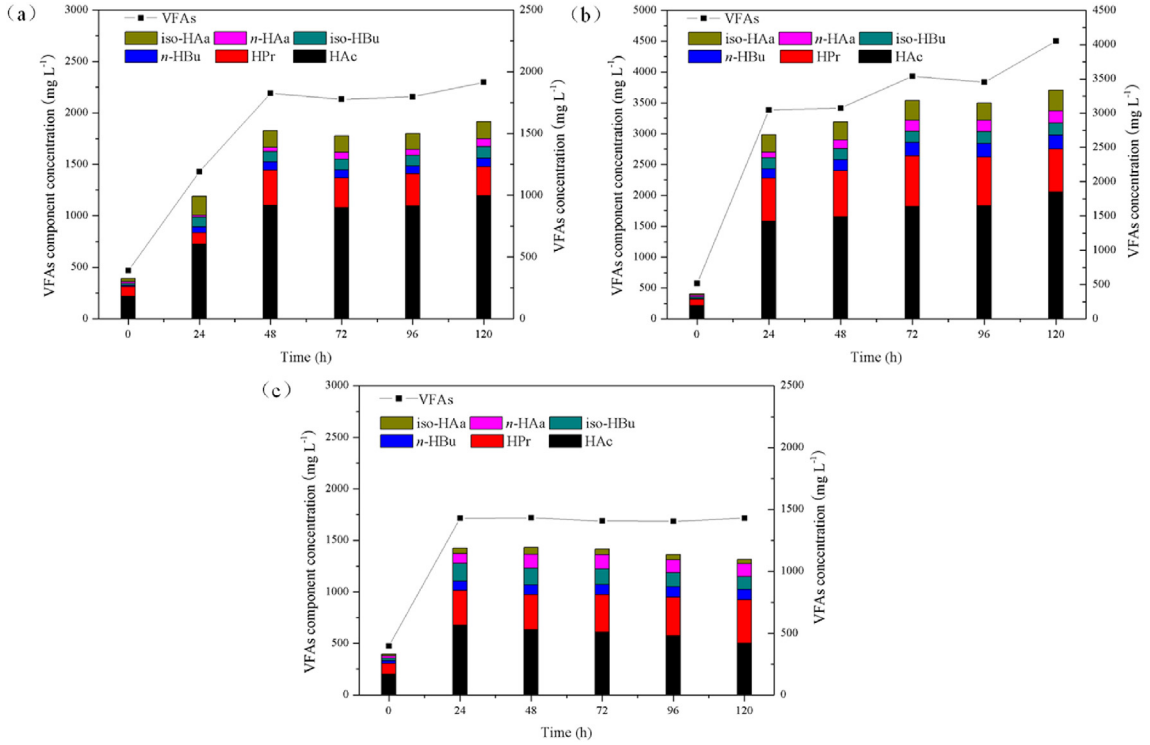


Fig. 2. Variation in VFAs concentration with (a) high-temperature micro-oxygen pretreatment (b) ultrasound-alkali pretreatment (c) alkali pretreatment.

temperature microaeration pretreatment can only be increased by 11 and 6 times, respectively. Ultrasound-alkali pretreatment is more favorable to disintegrate the WAS flocs and disrupt bacterial cell walls. After different pretreatments, the variation of VFAs showed a similar trend. The raise of VFAs accumulation of sludge pretreated by ultrasound-alkali was the biggest, which was increased by more than 6 times. By contrast, VFAs accumulation of sludge pretreated by other methods increased less. From the time perspective, the fermentation process can be kept stable in four or five days. It was found that the dissolution of soluble carbohydrate and protein can keep a balance with the hydrolysis and fermentation.

According to the experimental results, ultrasound-alkali pretreatment has prominent advantage over other pretreatment methods. Organic matters could be effectively released from the WAS pretreated by ultrasound-alkali. Therefore, ultrasound-alkali pretreatment was the most effective pretreatment method, it was used in the later experiments.

3.2. Methane-producing efficiency in MEC-AD coupled system

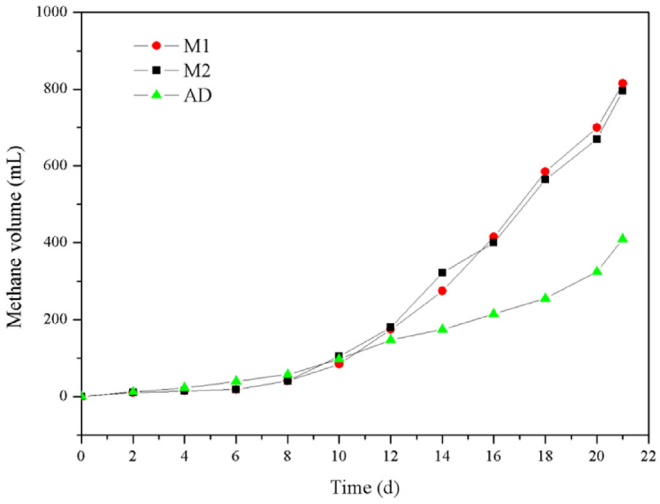


Fig. 3. Comparison of methane yields in different reactors.

When the bioelectrochemical system was operated, the difference between the MEC-AD and AD reactor in methane production has become increasingly apart. In general, the methane production in the coupled reactors was significantly higher than that in the AD reactor. As can be seen from Fig. 3, the accumulated methane barely increased in 0-5th day, without anode and applied voltage in coupled reactors. Carbon brush anode was put into MEC-AD coupled reactors on the fifth day, gas production had no obvious change within two or three days, which was due to the low current (less than 2 mA). It was reported that the carbon brush needed to adapt to the new environment just after it was placed in the reactors (De Vrieze et al., 2014; Liu et al., 2016; Sun et al., 2014). Compared to AD reactor, the methane production rate in the MEC-AD coupled reactors were increased faster after the eighth day. The

Table 2
Comparison of dissolved matters from WAS under different pretreatments.

Pretreatment method	Maximum VFAs concentration (mg L^{-1})	Maximum SCOD concentration (mg L^{-1})	Ultimate proteins concentration (mg L^{-1})	Ultimate carbohydrates concentration (mg L^{-1})
high-temperature microaeration	1915	3333	949	298
Ultrasound-alkali	4052	6872	274	134
Alkali	1432	2667	250	125

methane production rate of the MEC-AD and AD reactors were significantly different after the eleventh day. The maximum methane production rate of MEC-AD coupled reactor was 0.15 m^3 CH_4/m^3 reactor/d within 12–18th day, thus enhancing the rate by 3 times compared to control AD (0.05 m^3 CH_4/m^3 reactor/d). The current in MEC-AD coupled reactors were also higher than before at this stage, which were reached more than 10 mA. After 18 days, methane production rate of AD reactor was obviously improved and raised to 0.073 m^3 CH_4/m^3 reactor/d, but there was no significantly changed in MEC-AD coupled reactor (0.109 m^3 CH_4/m^3 reactor/d). Likely, the rate of methane production in MEC-AD coupled reactors will slow down as the concentration of organic substrate decreased.

The maximum accumulated methane yield of MEC-AD reactors was 808 ± 8 mL, which was significantly higher than the yield of AD reactor (410 mL). In other words, compared with AD reactor, the methane yield of MEC-AD reactors was improved by 97.0% ± 1.85%. It was found that hydrogenotrophic methanogens could use H_2 synthesized from cathode to convert CO_2 into methane in bioelectrochemical systems (Liu et al., 2010). The MEC-AD coupled system broke through the limitation that methane was produced only by acetotrophic methanogens. Hence, MEC-AD coupled system was proved to have great potential to enhance methane production rate and methane yield, with a current in excess of 10 mA (Liu et al., 2012c).

3.3. Degradation effect of WAS

3.3.1. Changes of VFAs concentration

VFAs began to accumulate when the anode was added to reactors M1 and M2 to form a circuit loop. The changes of VFAs were shown in Fig. 4, VFAs concentration in M1 and M2 had a similar trend. The concentration of VFAs was substantially increased in the first 11 days. At this stage, very little gas was detected, and the consumption of VFAs was low, thus resulting in a raise of VFAs accumulation. However, with the methane production rate and the consumption of VFAs increased, the VFAs accumulation was significantly decreased in MEC-AD coupled reactors after the twelfth day. In addition, the VFAs accumulation in control AD has been increasing from beginning to end. According to the calculation, VFAs concentration in reactors M1 and M2 were decreased by 582 and 811 mg L^{-1}, respectively. It should be emphasized that HAc content of total VFAs decreased from 53.5 ± 0.5% to 29.5 ± 0.3% in coupled reactors. Besides HAc, other acids were not changed significantly. This indicated that HAc was an important substrate in the process of anaerobic digestion, and it was always the first substrate to be used in the bioelectrochemistry process.

The reduction of HAc in reactors M1 and M2 were 730 and 879 mg L^{-1}, respectively, and their variation values were closed to those of VFAs. Therefore, the change of HAc concentration directly led to the change of the total VFAs concentration. According to the equation of methane production from HAc, 1 mol of HAc can theoretically produce 1 mol of methane. The theoretical methane yield in reactors M1 and M2 are 145 mL and 174 mL, respectively, in accordance with the decrease of HAc. But the actual methane accumulation was far greater than theoretical values, which was due to the HAc in the reactors were constantly being synthesized and consumed.

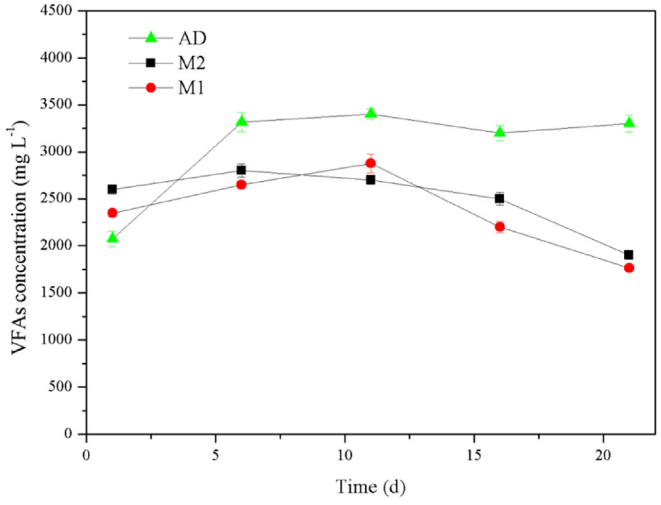


Fig. 4. Comparison of VFAs concentration in different reactors.

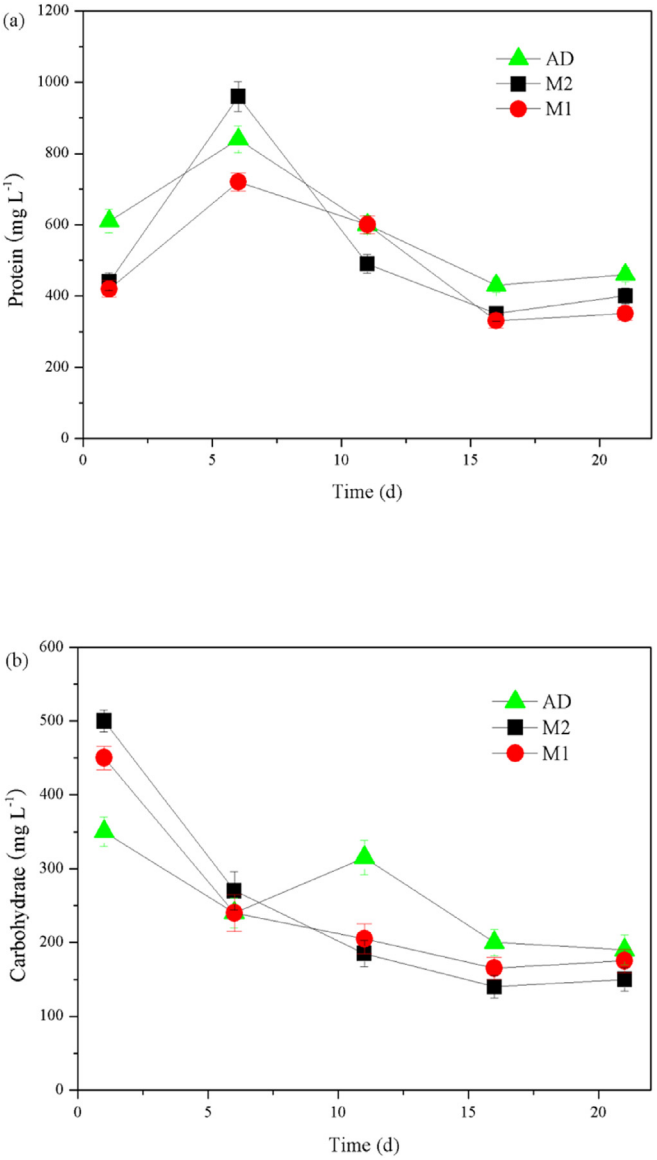


Fig. 5. (a) Changes of protein in different reactors; (b) Changes of carbohydrate in different reactors.

3.3.2. Consumption of soluble organics

Proteins and carbohydrates accounted for almost 70–80% of the main constituents of WAS. The concentrations of proteins in all reactors were increased at 0-5th days, and then it soon decreased (Fig. 5(a)). The initial concentration of proteins in reactors M1 and M2 were 412 and 439 mg L^{-1}, respectively, and it increased to 716 and 950 mg L^{-1} at the 5th day, respectively. The increase of protein concentration indicated that the rate of cell lysis and protein release was pretty fast at this stage. Subsequently, protein concentration was decreased. In contrast to the change of protein concentration, as shown in Fig. 5(b), the decreasing

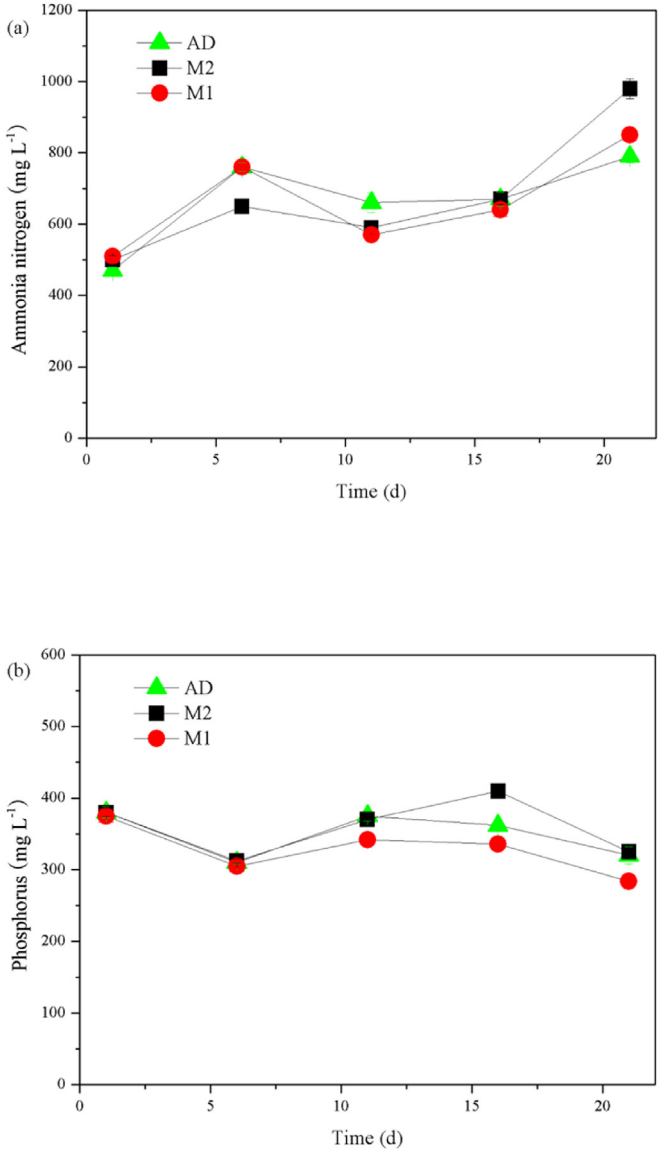


Fig. 6. (a) Changes of ammonia nitrogen in different reactors; (b) Changes of total phosphorus in different reactors.

trend of carbohydrate was more obvious. The concentration of carbohydrate in the MEC-AD coupled reactors has been decreasing all the time. Soluble protein and carbohydrate were rapidly oxidized in the anode, and the generated protons and electrons were combined to produce hydrogen in the cathode. At the same time, soluble organics were converted to HAc by acetogens in the reactors. HAc and hydrogen were used as substrate of methanogens to be further oxidized for producing methane. On the whole, the final concentration of protein and carbohydrate in MEC-AD coupled reactors were lower than those in AD reactor. The degradation of proteins and carbohydrates were enhanced in MEC-AD coupled system.

The main source of ammonia nitrogen was ammonification after proteolysis; thus, the change of ammonia nitrogen has a certain relation with the change of protein. The release rate of ammonia nitrogen increased with the raise of protein, therefore, the concentration of ammonia nitrogen also significantly increased in the first 5 days (Fig. 6(a)). After that, ammonia nitrogen was continuously accumulated with the process of proteolysis. As can be seen from Fig. 6(b), The phosphorus concentration during the whole system did not change much, which proved that it was always in equilibrium between phosphorus release and phosphorus absorption, and the variation was within the range of 100 mg L^{-1}. There was little difference in phosphorus changes between the M1, M2 and AD reactor at the first 12 days, but the difference occurred at the end of the fermentation (12-21 d), indicating that a different process has taken place. It appears that the fluctuation of phosphorus concentration may be affected by the metabolism of microorganisms.

Table 3
VSS and TCOD removal efficiency of sludge.

Reactor	VSS removal efficiency	TCOD removal efficiency
M1	19.36%	11.91%
M2	24.36%	15.30%
AD	3.71%	9.20%

3.4. VSS and TCOD removal efficiency

MEC-AD coupled reactors produced a lot of methane, so its VSS and TCOD removal efficiency had an obvious improvement compared with the AD reactor. As shown in Table 3, the removal efficiencies of TCOD in reactors M1 and M2 were 11.91%, and 15.30%, respectively, which were higher than that of AD reactor (9.20%). The TCOD removal efficiency of M2 was much higher than M1 due to the oxygen disturbance. Because a new anode was replaced in reactor M2 at 16th day, which resulted in the oxygen entering into the reactor. It was found that oxygen had no effect on methane production; however, it has a certain effect on the removal of TCOD. This may be because that the oxygen in reactors can be quickly consumed by the cathode biofilm and facultative anaerobes in the sludge under the effect of the applied voltage. However, this microaeration condition provided an appropriate reduction potential for TCOD removal. It was found that the entry of oxygen would leave the reactor being in an aerobic environment where the microorganisms were in the stage of endogenous respiration. Subsequently, the oxygen was consumed by the microorganisms, and oxygen content was reduced, then the reactor gradually entered the stage of microaeration. Due to the impact of oxygen, VSS removal efficiency of reactor M2 (24.36%) was also higher than that of reactor M1 (19.36%). Microaeration conditions help to stimulate cell lysis (Jenicek et al., 2014; Johansen and Bakke, 2006; Niu et al., 2016), and it can play a valid role in promoting the anaerobic digestion. Hence, the removal of VSS and TCOD were enhanced in MEC-AD coupled system.

4. Conclusions

The pretreatment of sludge can effectively disintegrate the WAS flocs and break the cell walls of the bacteria in WAS, and then WAS release organic matters. The maximum accumulated concentration of VFAs and SCOD in the WAS pretreated by ultrasound-alkali were excess 6.4-fold and 13.8-fold that of initial concentration, which were apparently higher than other pretreatment methods. Organic matters could be effectively released from the WAS pretreated by ultrasound-alkali. Therefore, ultrasound-alkali pretreatment was the most effective pretreatment method.

Under the premise of using ultrasound-alkali pretreatment, in fermentation experiments, a small amount of hydrogen could be detected in MEC-AD coupled reactors. The results illustrated that organic matters were oxidized by microorganisms at the anode, and the hydrogen was synthesized at the cathode. After that, hydrogen was used as the substrate of hydrogenotrophic methanogens to stimulate the production of methane. Meanwhile, organic compounds were converted to HAc by acetogens in the reactors. HAc and hydrogen were used as substrate of methanogens to be further oxidized for producing methane. Methane began to accumulate at about the fifth day. In MEC-AD coupled reactors, its bioelectrochemical enhanced effects became manifest at about the tenth day. The maximum rate of methane production can be

reached 0.15 m^3 CH_4/m^3 reactor/d in MEC-AD coupled reactors. Compared to AD reactor (0.05 m^3 CH_4/m^3 reactor/d), it was improved by 3 times. For methane yield, the yield of methane in MEC-AD coupled reactors can be achieved 815 and 800 mL, while that of AD reactor can only be achieved 410 mL. Methane yield of reactors M1 and M2 were respectively improved by 98.8% and 95.1% compared with AD reactor. The VSS removal efficiency was improved from 3.71% of AD to 19.36% and 24.36% of reactors M1 and M2, respectively. TCOD removal efficiency was enhanced from 9.20% of AD to 11.91% and 15.30% of reactors M1 and M2, respectively. It can be seen from the experimental results that the MEC-AD coupled system has a faster rate of utilizing the sludge substrate, the ultimate yield and rate of methane production were significantly improved. Compared with the simple AD reactor, the MEC-AD coupled system has more advantages in the energy conversion of WAS, and the addition of the bioelectrochemical system can assist the production of methane. MEC-AD coupled system has a good effect on improving the efficiency of methane production.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Appels, L., Baeyens, J., Degrève, J., Dewil, R., 2008. Principles and potential of the anaerobic digestion of waste-activated sludge. Prog. Energy Combust. Sci. 34, 755–781.
Ariunbaatar, J., Panico, A., Esposito, G., Pirozzi, F., Lens, P.N.L., 2014. Pretreatment methods to enhance anaerobic digestion of organic solid waste. Appl. Energy 123, 143–156.
Astals, S., Esteban-Gutiérrez, M., Fernández-Arévalo, T., Aymerich, E., García-Heras, J., Mata-Alvarez, J., 2013. Anaerobic digestion of seven different sewage sludges: a biodegradability and modelling study. Water Res. 47, 6033–6043.
Bo, T., Zhu, X., Zhang, L., Tao, Y., He, X., Li, D., Yan, Z., 2014. A new upgraded biogas production process: coupling microbial electrolysis cell and anaerobic digestion in single-chamber, barrel-shape stainless steel reactor. Electrochem. Commun. 45, 67–70.
De Vrieze, J., Gildemyn, S., Arends, J.B., Vanwonterghem, I., Verbeken, K., Boon, N., Verstraete, W., Tyson, G.W., Hennebel, T., Rabaey, K., 2014. Biomass retention on electrodes rather than electrical current enhances stability in anaerobic digestion. Water Res. 54, 211–221.
Guo, X., Liu, J., Xiao, B., 2013. Bioelectrochemical enhancement of hydrogen and methane production from the anaerobic digestion of sewage sludge in single-chamber membrane-free microbial electrolysis cells. Int. J. Hydrogen Energy 38, 1342–1347.
Hu, H., Fan, Y., Liu, H., 2009. Hydrogen production in single-chamber tubular microbial electrolysis cells using non-precious-metal catalysts. Int. J. Hydrogen Energy 34, 8535–8542.
Jenicek, P., Celis, C.A., Krayzelova, L., Anferova, N., Pokorna, D., 2014. Improving products of anaerobic sludge digestion by microaeration. Water Sci. Technol. 69, 803–809.
Jiang, J., Zhao, Q., Wang, K., Wei, L., Zhang, G., Zhang, J., 2010. Effect of ultrasonic and alkaline pretreatment on sludge degradation and electricity generation by microbial fuel cell. Water Sci. Technol. 61, 2915–2921.
Jin, L., Zhang, G., Tian, H., 2014. Current state of sewage treatment in China. Water Res. 66, 85–98.
Jin, Y.Y., Li, H., Mahar, R.B., Wang, Z.Y., Nie, Y.F., 2009. Combined alkaline and ultrasonic pretreatment of sludge before aerobic digestion. J. Environ. Sci. 21, 279–284.
Johansen, J.E., Bakke, R., 2006. Enhancing hydrolysis with microaeration. Water Sci. Technol. 53, 43–50.
Kargi, F., Catalkaya, E.C., Uzuncar, S., 2011. Hydrogen gas production from waste anaerobic sludge by electrohydrolysis: effects of applied DC voltage. Int. J. Hydrogen Energy 36, 2049–2056.
Lee, H.S., Vermaas, W.F., Rittmann, B.E., 2010. Biological hydrogen production: prospects and challenges. Trends Biotechnol. 28, 262–271.
Liang, D.W., Peng, S.K., Lu, S.F., Liu, Y.Y., Lan, F., Xiang, Y., 2011. Enhancement of hydrogen production in a single chamber microbial electrolysis cell through anode arrangement optimization. Bioresour. Technol. 102, 10881–10885.
Liu, W., Huang, S., Zhou, A., Zhou, G., Ren, N., Wang, A., Zhuang, G., 2012a. Hydrogen generation in microbial electrolysis cell feeding with fermentation liquid of waste activated sludge. Int. J. Hydrogen Energy 37, 13859–13864.
Liu, W.Z., He, Z.W., Yang, C.X., Zhou, A.J., Guo, Z.C., Liang, B., Varrone, C., Wang, A.J., 2016. Microbial network for waste activated sludge cascade utilization in an integrated system of microbial electrolysis and anaerobic fermentation. Biotechnol. Biofuels 9, 83.
Liu, W.Z., Huang, S.C., Zhou, A.J., Zhou, G.Y., Ren, N.Q., Wang, A.J., Zhuang, G.Q., 2012b. Hydrogen generation in microbial electrolysis cell feeding with fermentation liquid of waste activated sludge. Int. J. Hydrogen Energy 37, 13859–13864.
Liu, W.Z., Wang, A., Sun, D., Ren, N.Q., Zhang, Y.Q., Zhou, J.H., 2012c. Characterization of microbial communities during anode biofilm reformation in a two-chambered microbial electrolysis cell (MEC). J. Biotechnol. 157, 628–632.
Liu, W.Z., Wang, A.J., Cheng, S.A., Logan, B.E., Yu, H., Deng, Y., Van Nostrand, J.D., Wu, L.Y., He, Z.L., Zhou, J.Z., 2010. Geochip-based functional gene analysis of anodophilic communities in microbial electrolysis cells under different operational modes. Environ. Sci. Technol. 44, 7729–7735.
Logan, B.E., 2005. Simultaneous wastewater treatment and biological electricity generation. Water Sci. Technol. 52, 31–37.
Lu, L., Xing, D., Ren, N., 2012b. Bioreactor performance and quantitative analysis of methanogenic and bacterial community dynamics in microbial electrolysis cells during large temperature fluctuations. Environ. Sci. Technol. 46, 6874–6881.
Luo, H., Liu, G., Zhang, R., Bai, Y., Fu, S., Hou, Y., 2014. Heavy metal recovery combined with H2 production from artificial acid mine drainage using the microbial electrolysis cell. J. Hazard Mater. 270, 153–159.
Nancharaiah, Y., Mohan, S.V., Lens, P., 2015. Metals removal and recovery in bioelectrochemical systems: a review. Bioresour. Technol. 195, 102–114.
Niu, T.H., Zhou, Z., Shen, X.L., Qiao, W.M., Jiang, L.M., Pan, W., Zhou, J.J., 2016. Effects of dissolved oxygen on performance and microbial community structure in a microaerobic hydrolysis sludge in situ reduction process. Water Res. 90, 369–377.
Pham, T., Rabaey, K., Aelterman, P., Clauwaert, P., De Schamphelaire, L., Boon, N., Verstraete, W., 2006. Microbial fuel cells in relation to conventional anaerobic digestion technology. Eng. Life Sci. 6, 285–292.
Rozendal, R.A., Jeremiasse, A.W., Hamelers, H.V., Buisman, C.J., 2007. Hydrogen production with a microbial biocathode. Environ. Sci. Technol. 42, 629–634.
Schröder, U., Harnisch, F., Angenent, L.T., 2015. Microbial electrochemistry and technology: terminology and classification. Energy Environ. Sci. 8, 513–519.
Selembo, P.A., Merrill, M.D., Logan, B.E., 2010. Hydrogen production with nickel powder cathode catalysts in microbial electrolysis cells. Int. J. Hydrogen Energy 35, 428–437.
Selembo, P.A., Perez, J.M., Lloyd, W.A., Logan, B.E., 2009. High hydrogen production from glycerol or glucose by electrohydrogenesis using microbial electrolysis cells. Int. J. Hydrogen Energy 34, 5373–5381.
Singh, A., Olsen, S.I., Pant, D., 2013. Importance of life cycle assessment of renewable energy sources. In: Life Cycle Assessment of Renewable Energy Sources. Springer, pp. 1–11.
Sun, R., Xing, D., Jia, J., Liu, Q., Zhou, A., Bai, S., Ren, N., 2014. Optimization of high-solid waste activated sludge concentration for hydrogen production in microbial electrolysis cells and microbial community diversity analysis. Int. J. Hydrogen Energy 39, 19912–19920.
Villano, M., Aulenta, F., Majone, M., 2012. Perspectives of biofuels production from renewable resources with bioelectrochemical systems. Asia Pac. J. Chem. Eng. 7, S263–S274.
Villano, M., Monaco, G., Aulenta, F., Majone, M., 2011. Electrochemically assisted methane production in a biofilm reactor. J. Power Sources 196, 9467–9472.
Wahidunnabi, A.K., Eskicioglu, C., 2014. High pressure homogenization and two-phased anaerobic digestion for enhanced biogas conversion from municipal waste sludge. Water Res. 66, 430–446.
Wang, L., He, Z., Guo, Z., Sangeetha, T., 2019. Microbial community development on different cathode metals in a bioelectrolysis enhanced methane production system. J. Power Sources 444 227306-227306.
Wang, L., Liu, W., He, Z., Guo, Z., Zhou, A., Wang, A., 2017. Cathodic hydrogen recovery and methane conversion using Pt coating 3D nickel foam instead of Pt-carbon cloth in microbial electrolysis cells. Int. J. Hydrogen Energy 42, 19604–19610.
Wang, L., Liu, W., Kang, L., Yang, C., Zhou, A., Wang, A., 2014. Enhanced biohydrogen production from waste activated sludge in combined strategy of chemical pretreatment and microbial electrolysis. Int. J. Hydrogen Energy 39, 11913–11919.
Wang, W., Luo, Y., Qiao, W., 2010. Possible solutions for sludge dewatering in China. Front. Environ. Sci. Eng. 4, 102–107.
Xiao, B., Yang, F., Liu, J., 2011. Enhancing simultaneous electricity production and reduction of sewage sludge in two-chamber MFC by aerobic sludge digestion and sludge pretreatments. J. Hazard Mater. 189, 444–449.
Xu, L.J., Liu, W.Z., Wu, Y.N., Wang, A.J., Li, S., Ji, W., 2013. Optimizing external voltage for enhanced energy recovery from sludge fermentation liquid in microbial electrolysis cell. International Int J Hydrogen Energy 38, 15801–15806.
Zhang, Y., Angelidaki, I., 2014. Microbial electrolysis cells turning to be versatile technology: recent advances and future challenges. Water Res. 56, 11–25.
Zhou, A., Yang, C., Guo, Z., Hou, Y., Liu, W., Wang, A., 2013. Volatile fatty acids accumulation and rhamnolipid generation in situ from waste activated sludge fermentation stimulated by external rhamnolipid addition. Biochem. Eng. J. 77, 240–245.
Zhou, A.J., Du, J.W., Varrone, C., Wang, Y.Z., Wang, A.J., Liu, W.Z., 2014a. VFAs bioproduction from waste activated sludge by coupling pretreatments with Agaricus bisporus substrates conditioning. Process Biochem. 49, 283–289.
Zhou, X., Jiang, G.M., Wang, Q.L., Yuan, Z.G., 2014b. A review on sludge conditioning by sludge pre-treatment with a focus on advanced oxidation. RSC Adv. 4, 50644–50652.